



Advanced Digital I&C Technology in Nuclear Power Plants A Success Story from Finland and China

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ABSTRACT

FRAMATOME ANP together with **Siemens PGL** have just commissioned the I&C systems for the Chinese Nuclear Power Plant Tianwan Unit 1, design the I&C systems for the first European Pressurized Water Reactor Olkiluoto 3 in Finland and started one of the most extensive and ambitious modernization project in a nuclear power plant for Loviisa NPP in Finland. The presentation delivers an insight into the problematic of digital I&C systems integration and focuses on aspects regarding design criteria, Structure and test approaches.

1 INTRODUCTION

In December 2003 the Finnish utility TVO awarded the Consortium FRAMATOME ANP/Siemens AG the contract for the turnkey construction of a nuclear power plant at the Olkiluoto site in Finland. For the first time, the newly developed EPR (European Pressurized Water Reactor) will be installed. I&C will be based on the advanced digital platform TELEPERM XS and TELEPERM XP which includes the Operating and Monitoring System OM 690. TXS, developed by ANP, together with TXP, developed by Siemens, constitutes a platform which has been proven in operation in the course of numerous modernization projects in Pressurized and Boiling Water Reactors from various suppliers worldwide. In the NPPs Tianwan 1&2, for the first time in China, a new NPP was completely automated with this technology.

Selection of the right instrumentation and control (I&C) system has a decisive influence on nuclear power plant safety and availability. These aspects therefore were particularly taken into account during the development of the I&C systems TELEPERM XS and TELEPERM XP.

Together, these two systems cover the entire range of automation applications in nuclear power plants.

- TELEPERM XP is primarily oriented to automation of the non-safety-related part of the power plant process. Such applications involve extensive open and closed-loop control systems and encompass all tasks required for process control via the man-machine interface.

The architecture of the TELEPERM XP process control system has been tailored to the process engineering needs of modern power plants.

By organizing the automation systems to match the process organization and by the hierarchical organization of the I&C system, the interaction of the individual components, devices and systems is supported.

In terms of operating convenience, the process control system TELEPERM XP is organized into hierarchical levels. This reduces the workload on the control room staff, allowing them to concentrate on their main tasks. A particularly ergonomic control room design ensures ease of operation.

- TELEPERM XS is designed to meet all the requirements on I&C important to safety in nuclear power plants. Typical applications include reactor protection (RPS) to initiate Reactor Trip and Engineered Safety Features Actuation System actuations (ESFAS).

TELEPERM XS is qualified for application in the highest safety category to be suitable for all of these tasks (eg: cat A of IEEE1226), and the scalability of the system makes it an equally cost-effective solution for other functions important to safety such as reactor control or use as control rod worth minimizer.

Specific requirements for safety I&C systems are defined in national and international codes and standards. These relate particularly to fault tolerance, robustness and qualification.

2 ADVANCED I&C CONCEPT FOR OLKILUOTO 3

2.1 General and Contractual Form

In December 2003 after nearly 15 years the first contract to supply a turnkey Nuclear Power Plant was signed in Europe. The investor and utility is TVO in Finland. The new plant will be erected in Olkiluoto, a location in the southwest of Finland where two Boiling Water Reactors have already been in operation for about 25 years. As reactor type the customer selected the European Pressurized Water Reactor, developed by FRAMATOME and Siemens in the late 1990's. The contractor is a consortium composed of FRAMATOME ANP and Siemens AG. ANP is the consortium leader and responsible for the Nuclear Island and related construction work which includes the NSSS, the Balance of Nuclear Island, Electrical Systems and I&C, the first Fuel Load, the Simulator, the overall coordination and the functional and technological Integration of the entire planning. Siemens supplies the Turbine Island and related construction work, which covers the Turbine, the Generator and the Balance of Turbine Island.

2.2 Fundamental Design Criteria

The most important characteristics of the EPR are: Core thermal power 4.300MW_{th}, Net power output 1577MW_{el}, Net efficiency 37%, four loops, four steam generators and double shell containment.

The EPR design philosophy is governed by three essential targets:

- Improving the safety level compared to existing plants by deterministic and probabilistic considerations
- Mitigation of hypothetical severe accidents by restricting their consequences to the plant itself
- Economic power generation costs competitive with other primary energy sources

The design criteria for OL3 result from several Finnish YVL guidelines. The most important technical requirements regarding I&C contain a defense in depth concept, diversity to cope with Common Cause Failure, degree of redundancy together with consideration of the I&C system response time, reliability targets and human factors. Following further attention will be paid on the Defense in Depth Concept. Defense in Depth exists of a set of staggered physical barriers like fuel cladding, reactor coolant boundary and containment isolation. A set of appropriate design provisions and operating strategies to protect the physical barriers are added:

- classification of functions, systems and structures according to their importance to safety
- deterministic design requirements according to safety class regarding redundancy, diversity, material quality, ...
- probabilistic design goals to balance the frequency of occurrence against potential consequences

The following lines of defense are determined:

- Preventive line of defense: Prevent departures from normal operation by conservative design, quality, assurance, surveillance of closed and open loop controls etc.
- Main line of defense: Mitigate accidents and prevent their evolution into severe accidents by engineered safety features and protective systems
- Risk reduction line: Cope with the complete loss of engineered safety features and protective systems by risk reduction functions
- Ultimate line of defense: Preserve the integrity of the containment in the case of severe accidents by special core melt retention and cooling devices.

Considering these lines of defense and the related Finnish regulatory guide YVL 2.7 the related I&C criteria are defined as following:

- Implementation of safety class 2 (SC 2) functions in a N+2 channel architecture, self monitored or periodic tested, example reactor protection system
- Implementation of safety class 3 (SC 3) functions in a N+1 channel architecture, self monitored or periodic tested, example rod control system
- Implementation of safety class 4 (SC 4) functions which have to be in general single failure tolerant, example diverse protection system

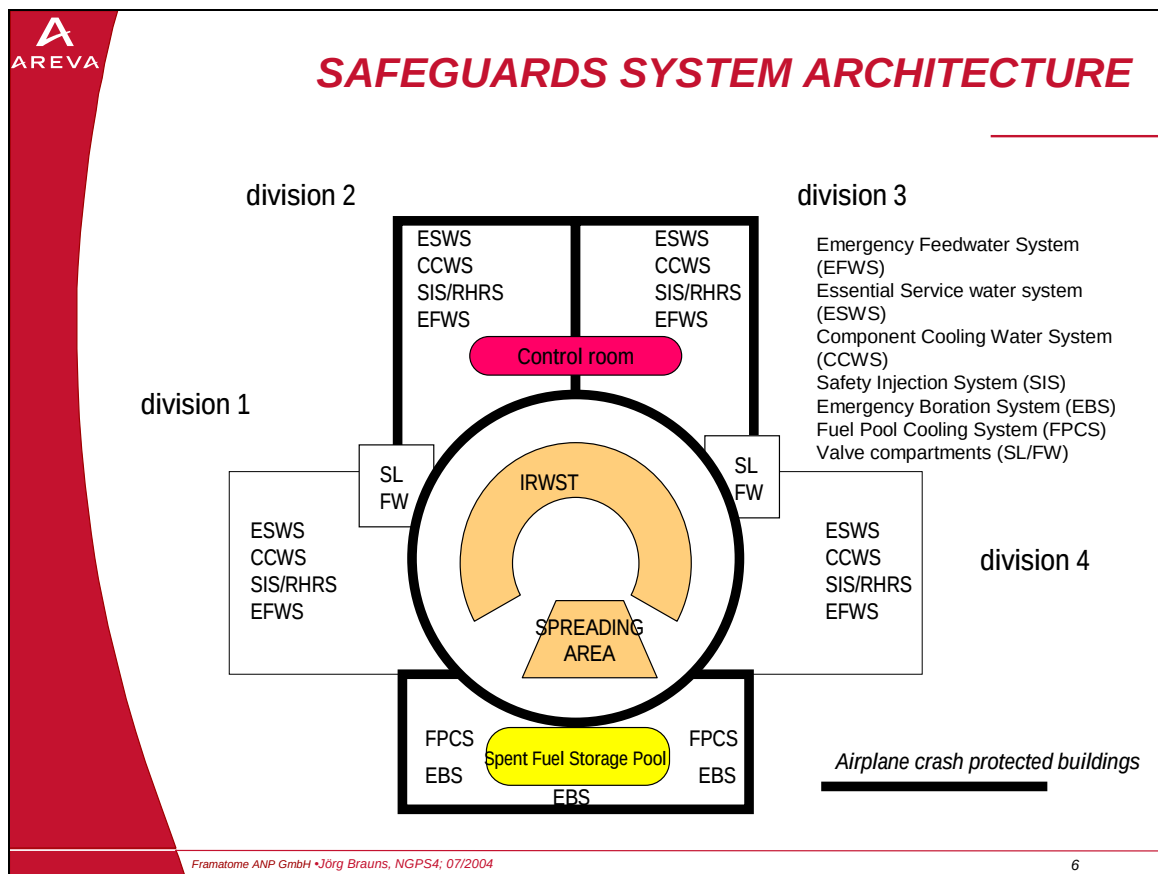
Complementing these definitions the non safety class NC is defined which includes all other I&C functions including operating and monitoring.

YVL guide 5.5 requires that all instrumentation and control needed to monitor and control the plant in severe accident condition has to be independent from other systems of the plant.

Last line of defense to be considered in the design of the OL3 plant is the loss of all computerized systems, for which is required the possibility to operate the plant up to a safe state, even in case of the occurrence of a frequent incident or accident.

2.3 I&C Design and Overall Architecture

The I&C design has to consider the above mentioned and partly described criteria. Main criteria are deterministic and deal with redundancy, independence and diversity.



Safeguard system Architecture

EPR design comprises four redundant safeguard buildings and safety related systems are four times redundant. Redundancy on I&C is driven from the mechanical systems themselves so the overall architecture is four times redundant ensuring a high reliability, particularly for the highest classified functions remaining operable even in case of a single failure superposed with corrective or preventive maintenance.

Independence is considered by allocating functions in different I&C systems in order to mitigate consequences in case one I&C system is postulated out of service (eg two sub-systems are implemented within the protection system; severe accident I&C is apart). Therefore, interfaces between independent systems are subject to decoupling measures to prevent failure propagation.

Diversity approach is subdivided into product approach and the diversity of the application profile.

Product diversity applies to avoid common design or manufacturing defects between components exposed to the same application profile.

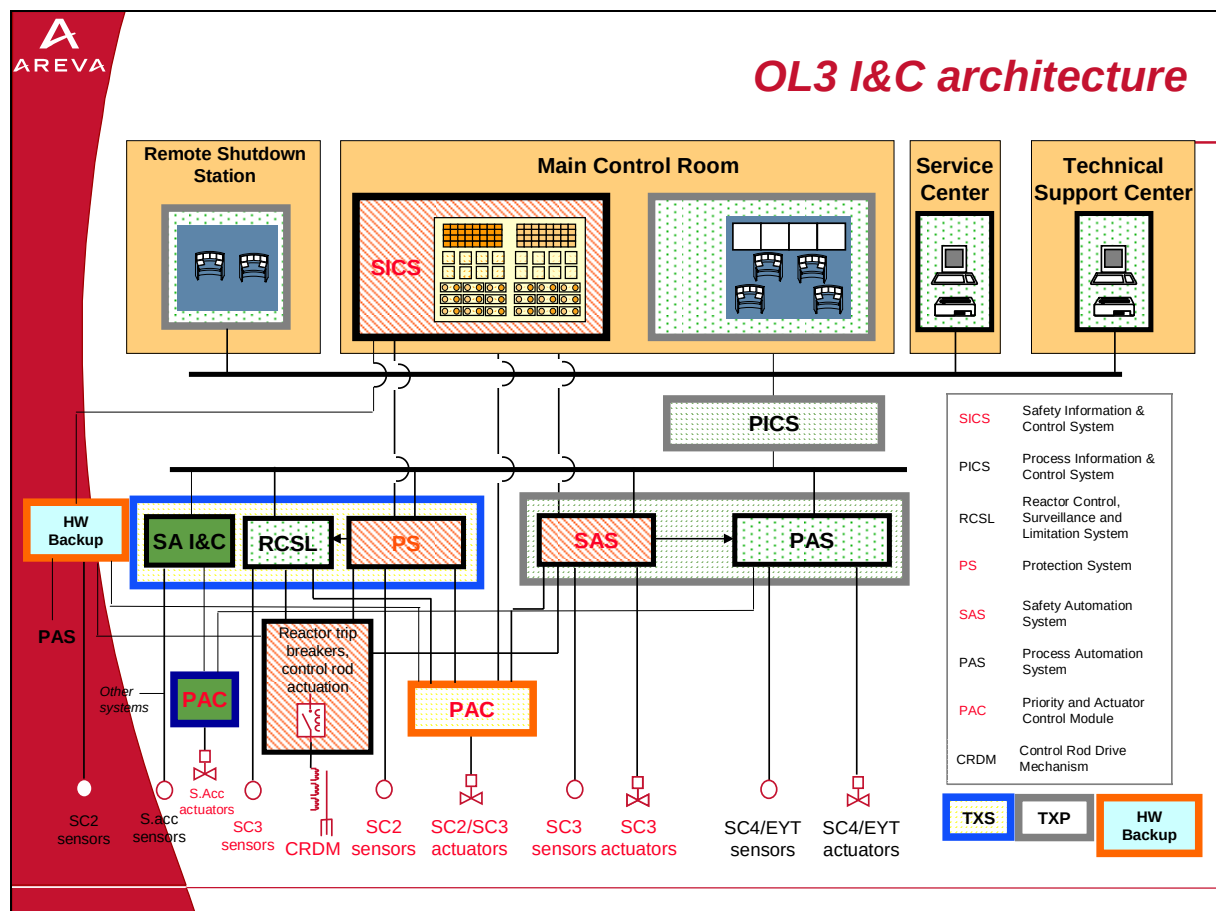
Diversity is considered at both automation level and Human Machine Interface level.

At the automation level, diversity between TELEPERM XS (TXS) and TELEPERM XP (TXP) is claimed to ensure the independence between Safety Automation System (SAS) based on TXP and the Protection System (PS) based on TXS. Components used for TXS and TXP are completely different. This is valid for all parts like the standard sub racks, the CPU's, the central I/O modules and the embedded controllers. Also the factories are different.

In addition, with regard to the loss of all computerized systems requirement, a third diverse technology, non computerized, is provided.

Regarding the HMI level two diverse information systems are available: The computer based Process and Information system (PICS) using TXP and the Safety Information and Control System using conventional pushbuttons and indicators.

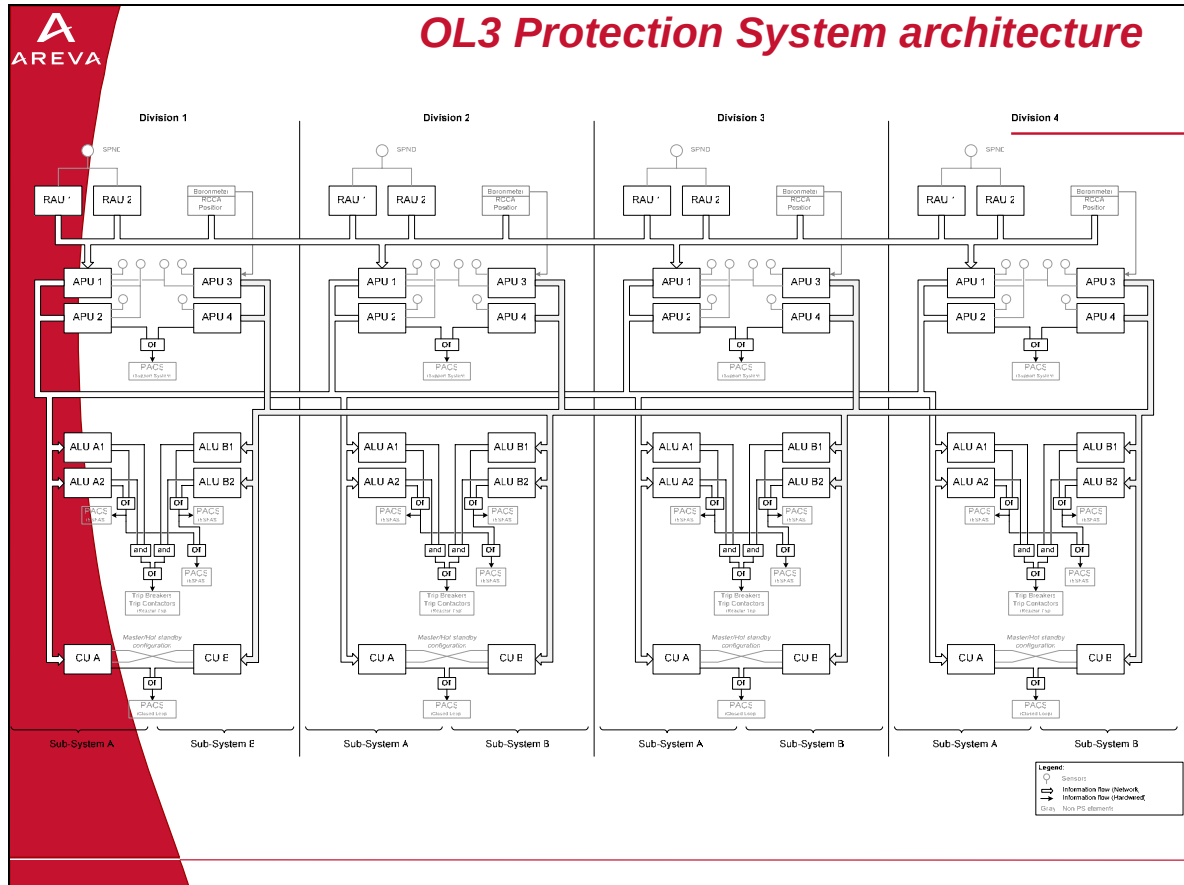
The diversity of the application profile applies to avoid a common triggering event between components of the same design. An example is the independence between Reactor Control, Surveillance and Limitation System (RCSL) and the PS as well as between the two subsystems (Diversity Group A and B) of the PS.



I&C General Design

The TXS/TXP platform completely fulfils all requirements. It is designed for comprehensive automation tasks in NPPs. The architecture is tailored according to specific

needs such as degree of redundancy and number of automation tasks. For instance, the Protection System applied for the highest I&C safety class 2 is designed with 4 redundancies from the sensors to the actuators. Additionally all reactor trip initiation functions are based on two diversity groups of physical parameters and processed in two independent parts of the protection system, TXS based



Protection System architecture overview

The safety I&C system TXS performs its tasks with a cycle time downward up to 5 ms, with fast communication links and by means of cyclic processing. Other general characteristics are comprehensive self monitoring capabilities via software and hardware and an outstanding degree of reliability. TXS is qualified in accordance with international safety rules, is NRC approved and used mainly for YVL safety class 2 applications. TXP is qualified for category B safety functions according to IEC 1226 and is used partly for YVL SC3 and for all SC4 and NC applications.

2.4 Priority Control

The Priority Actuation Control is used when different safety classified actuation signals or when safety classified and not safety classified actuation signals are addressed to the same electrical actuators. It is specially developed and safety classified as part of the safety I&C system TXS. Per electric actuator one PAC module is required.

The PAC module consists of SC2 conventional components (like input output circles), SC2 programmable logic device-PLD for SC2 functions and a micro processor with serial links for SC3 and SC4 functions.

The PAC module has been qualified according the German KTA 3505 and assessed by the German TÜV for the application in the PS. The main items of qualification have been:

- The PLD functions have been completely tested including the conventional components
- The possible impact of the SC3 components and functions on the SC2 part was tested and assessed to be sufficiently low, because of the implemented functional isolation measurements. Moreover, SC3 components can be “disconnected” keeping only operable the SC2 PLD functions.
- The hardware of the PLC module (SC2 and SC3) is qualified according to the environmental requirements.
- The interface in the switchgear consists of relays for the on/off-commands and isolated (potential-free) contacts

2.5 Screen based Main Control Room

The screen based HMI (Human Machine Interface) OM690 provides the full benefits of the digital I&C platform. Main features are rapid and guided access to related information and diagnostics, task oriented information, support of situation awareness, integration of information, control and documentation, easy and rapid modifications.

Operation of the plant from the main control room is performed by using screen based HMI in all plant situation.

It consists of four operator workplaces with 5 screens each. Two workplaces are in operating mode, enabling to control components as the two others are in supervision mode. In case of failure of one operator workplace, others can be reconfigured in operation mode to ensure a high availability of the HMI computerized system.

OM690 computerized HMI provides suitable features for the screen based operation of Nuclear Power Plants:

- Plant displays are organized in hierarchy that ensures a simple and fast access to information the operator need.
- Alarm system provides instant access to plant displays where the operator will find all needed information and controls to perform required actions. Moreover, alarms are filtered according to plant situation criteria.
- Field data are all archived and accessible by operators under the form of lists and trend curves.
- Operating procedures as well as alarm sheets are accessible directly from plant displays and/or alarm lists.

In addition to the four operator workplaces, a Plant Overview Panel (POP), consisting of four large screens, provides operators overall Plant synoptic and field data. Similar to operator workplaces, operators can select displays to be activated on the POP.

Failure of the HMI computerized system is part of the EPR design, so diverse information and control devices are available in the main control room. Safety Information and Control System (SICS) consists in a set of discrete components (desk tiles, indicator) and

qualified screen based displays all diverse from the OM690 platform. SICS design ensures plant operation in post accident conditions.

EPR design comprises also a Remote Shutdown Station, geographically, used in case of unavailability of the main control room such as a fire in MCR. It consists of two operator workplaces with a limited number of screens. Same features as for the main operator workplaces are available.

2.6 Test concept for the safety I&C system TXS

The validation process of the safety I&C system TXS (SC2) is taking feedback of experience of previous projects.

Improvements are focused on the test coverage that will be quantified and verified by means of tools. Validation process of safety functions will be performed as summarized below by the independent Verification and Validation team (V&V):

- Based on I&C function specifications, V&V is developing a software Oracle. This Oracle is instrumented according to project rules in order to check, when test cases will be run, that all configurations that must be tested to get an adequate coverage of the design are effectively covered by the test cases set.
Test cases set are adjusted until the test coverage criteria are met.
- Test cases set are then executed on the final software designed by the production team, by the independent V&V team by using the SIVAT TXS simulator.
- Once the previous phase is successful, global tests are performed on TXS platform before site delivery.

3 I&C SYSTEM FOR ENTIRE TIANWAN NPP IN CHINA

3.1 General and Contractual Form

The Tianwan nuclear power plant consists of 2 units with Russian pressurized water reactors of the type VVER1000 / 428. The thermal power per unit amounts to 3000 MW(th) and the electrical power is 1070 MW. Type VVER 1000 consists of four loops and horizontal steam generators. The utility is the Jiangsu Nuclear Power Corporation. The plant is located in the Province Jiangsu between Beijing and Shanghai. The designer is the Saint Petersburg Research and Design Institute "ATOMENERGOPROJECT". The main I&C part was supplied by AREVA (FRAMATOME ANP) and Siemens. The first unit will commence commercial operation in 2005, the second unit in 2006. The I&C System in unit 1 has already been commissioned and is ready for operation. The license for the I&C is based on Chinese Nuclear standards.

JNPC decided to equip TNPP exclusively with digital I&C as the first Nuclear Power Plant worldwide. TELEPERM XS/XP has been considered as the most modern and also approved digital I&C system for safety and non safety applications in NPPs. The system has been qualified to according the safety requirements of the Chinese authority and fulfills the performance requirements of both the Chinese customer and the Russian plant designer. Another advantage is that AREVA (FRAMATOME) has experience with the design of Russian VVER power plants.

The I&C project is based on a trilateral frame contract between the Chinese customer, the Russian party and the German party which consists of the consortium Siemens-FANP, and a procurement contract between the Chinese customer and Siemens-FANP. The I&C responsibilities of the three parties are divided as follows:

The Russian party has the general technical responsibility for the construction of the plant, the responsibility for functional and technical requirements, for the input data for the I&C, for the process equipment performance algorithms and for the process technical requirements.

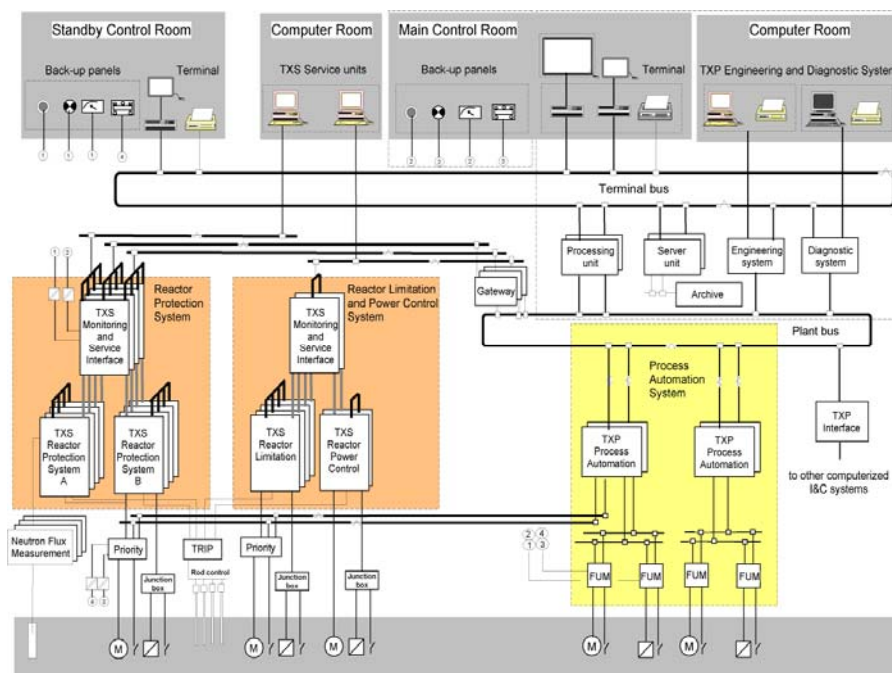
The Chinese party took over the co-ordination, the licensing, the approvals and provided the initial site data.

The German consortium Siemens-FANP has the technical responsibility for the I&C system and the I&C management for engineering, supplies, manufacture, testing, erection, commissioning, further-more the responsibility for the interface management of third-party I&C systems.

The I&C structure consists of the HMI (Human Machine Interface) which is performed with screens and panels in the main control room and the emergency control room, the non-safety I&C scope for start up, power operation and normal shutdown operation and the safety I&C scope which deals with abnormal plant situations. The computer-based operating and monitoring system OM-690 for the HMI contains 6 Operation Terminals (OT) with 18 working displays and 2 large screen displays, the Processing Units (PU) and the Server Units (SU). The processing of the information shown on the operator screens is performed by the PUs. The SUs mainly provide the archiving system. The OTs are responsible for serving the Operator Screens as well as the Large Screens for plant overview with graphics. The OM 690 serves among other tasks as interface to operate the plant manually. Functions important to safety are implemented in addition on independent back-up panels.

The communication between the PUs, SUs and OTs is realised via a fibre optic Terminal Bus. This permits the location of OT's with screens also locally e.g. in the SCR (Standby Control Room). Information from foreign systems to be connected with OM690 can be exchanged via the XU units.

Tianwan 1&2: I&C Structure



Advanced digital I&C technology for Olkiluoto 3 and Tianwan 1&2

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Overall I&C

The reactor protection system (RPS) is performed by the 1E qualified TXS platform and consists of two functionally diverse sub-systems A and B, each with four redundancies. To cope with possible common cause failures (CCF) two physically diverse criteria are applied for each design based accident. The signal exchange between redundancies for signal validation or voting is realised by optical fibres to ensure electrical and functional isolation. The RPS interacts with the reactor instrumentation and actuation systems (e.g. in-core and ex-core neutron flux measurement, power breaker system, rod control system etc.) which are under the Russian scope of supply. Other process parameters such as temperature, pressure, flow rate etc. are fed into TXS directly via Signal Conditioning ("SC") modules. The RPS also has connections with 1E qualified reactor safety panels in the main and standby control rooms (MCS and SCR) partly via direct I/Os and partly via the so called Monitoring & Service Interface ("MSI") computers. MSI computers also connect the Service Unit in the I&C service centre. The service unit is used for engineering and diagnosis purposes and for conducting periodic tests of the TXS system. MSI computers connect additionally the RPS to the operational I&C TXP via redundant Gateway computers to enable unidirectional signal exchange from TXS to TXP. The 1E safety drives are connected to TXS via the Priority Logic modules AV42. AV42 also receives manual commands from the safety panels in the control rooms as well as control signals from TXP (via a fibre optic bus). The priority logic ensures that the automatic protection commands from RPS always have higher priority over possible manual or automatic commands from operational (NS) I&C.

Safety related functions are realised partly in TXS and partly in TXP. The closed loop control for reactor power is SR classified in TNPP. The function is realised in TXS in a

master/slave structure with hot standby. TNPP has additionally automated reactor power limitation functions classified as safety related (high) category. They provide automatic power cutback or prohibit the power increase to minimise the demand on reactor protection actions. Thus the safety and availability of the plant is increased. The reactor power limitation system has a four redundancy structure, but without functional diversity requirements.

Typical SR functions realised in TXP are important closed loop controllers e.g. control of emergency feed water supply, main steam relief control etc. Another type of SR functions is control of ventilation systems for rooms which are important to safety but normally 4 train redundant. Single failure in such a system will have negative impacts on plant safety only in the long term.

TXP is qualified for such SR tasks due to the fact that it has a controlled and documented design and development process for hardware and software on the one hand and enough operational experience in more than 1000 applications (objects) in nuclear or fossil power plants on the other. In TNPP mainly the automation system AS620B of TXP family is used. The hardware of this basic automation system is type tested according to IEC1131-2.

Additional environmental qualification like long term (dry heat) and seismic tests have been performed according to IEEE344 to demonstrate the fulfilment of requirements for functionality during and after an earthquake in nuclear power plants.

AS 620B is typically configured as automation processor (AP) pairs in hot standby with redundant bus connection to the function modules (FUM). Input signals from measurement or output signals to actuators are wired directly to FUMs. The APs carry out the control and protection logic and are connected redundantly to the fibre optic Plant Bus. The plant bus is based on the highly available SIMATIC NET bus system and is single failure tolerant. It is used for communication between the automation systems AS620B or AS620T (for turbine control), the engineering system ES680, the diagnosis system DS670 and the operation and monitoring system OM690.

The scope of supply contains the automation of all technical processes, the measuring transducers, the operating terminals and back-up panels in the main control room and the emergency control rooms, the reactor protection system for trip and engineered safety features functions, the reactor limitation system and the automatic power controller. Here are some figures to quantify the biggest project which ever has been performed with TELEPRM XS/XP:

- 67 TELEPERM XP cabinets
- 64 TELEPERM XP cabinets
- approx. 3000 Binary inputs
- approx. 6000 Analog inputs
- approx 3400 drives and
- thereof 628 with priority control

Regarding the TXS, further challenges were the first application of a successor central processor SVE2, the newly developed priority module AV 42 and new testing peripherals called ERBUS.

3.2 Test concept for the safety I&C system TXS

The verification & validation process of the safety I&C system TXS was distinctively improved in the scope of the execution of the Tianwan project.

All plant specific tests to verify the requisite quality and proper functioning of the TELEPERM XS safety I&C contain a sequence of test activities. Prerequisite for starting the next test is successful completion of the previous one.

The test sequence for TXS safety I&C is subdivided into different test phases:

A: Test of application software

The validation of the application software was realized in two main steps. Firstly a so called “white box test” was performed in AREVA’s (FRAMATOME ANP) I&C department. Plant specific safety functions were performed with the aid of special validation tools. Secondly a “black box” test was executed by the Russian process engineers. All open loops and closed loops were tested to validate the system behavior during the simulation of Anticipated Operational Occurrences (AOO) and Design Based Accidents (DBA).

B: System test

Main objective is the verification of safety I&C specification in the test bay. All relevant parts of the TXS system, including the interfaces formed by the gateways and reactor protection panels, are set up and an integral function test performed on these following implementation of the software tested in phase A on the hardware.

A test machine is used to simulate all analog and binary input signals and collect all output signals from TXS system.

During the Factory Acceptance Test (FAT) the required system performance and failure behavior as well as the compliance with the functionality, defined in technological task description are demonstrated.

C: Commissioning of TXS system

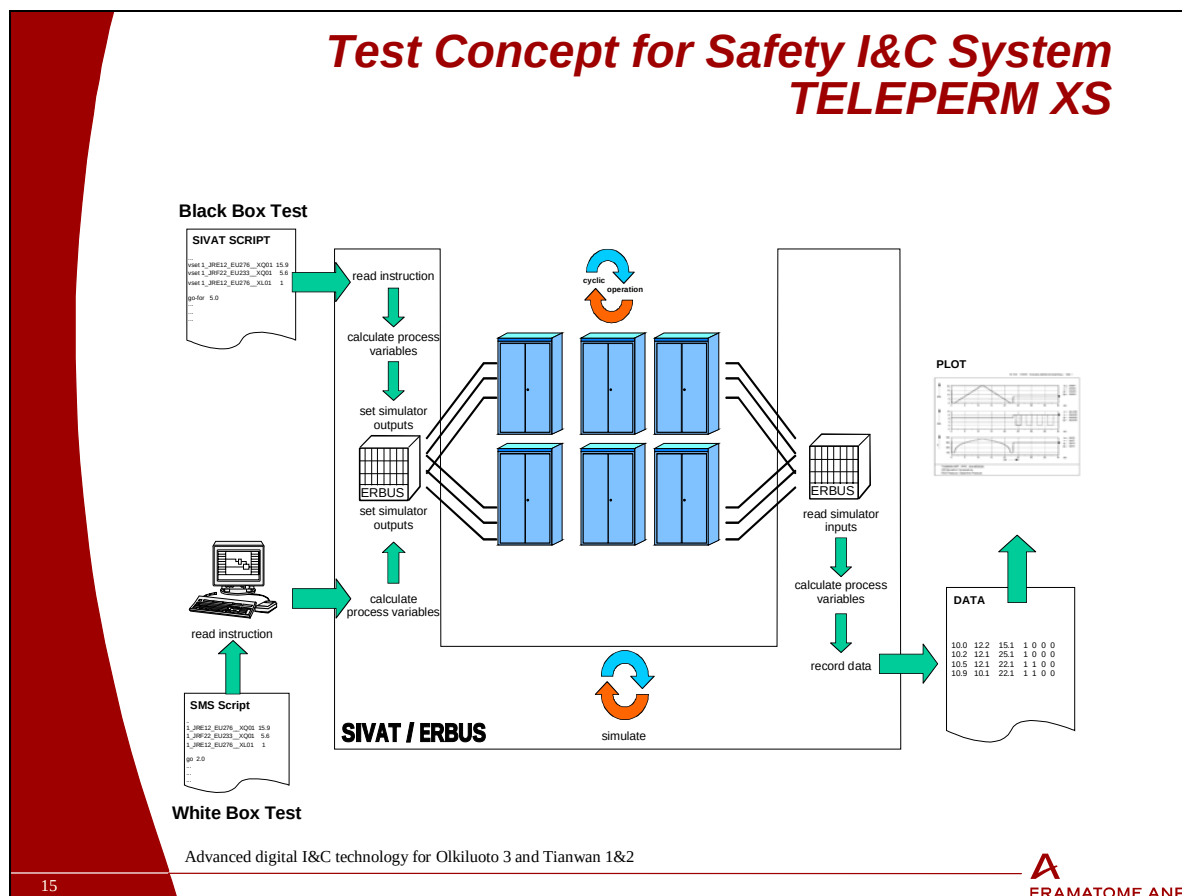
Main objective is the verification of the specification of the TXS safety I&C installed in the plant. Prerequisites for this are operability of the power supply and completion of assembly of the TXS systems, as well as of the other I&C systems required for overall function.

After successful cold commissioning the TXS safety I&C is ready for take over to temporary operation (TOTO).

D: Plant commissioning

Main objective is to verify that interaction of the plant components and of the TXS safety I&C under various operating and accident conditions and disturbances and in all relevant plant conditions is as designed. This is achieved by performing plant tests on start-up and in power operation. The parameters are optimized by evaluation of the dynamic plant tests.

As a result of this comprehensive and overlapping test procedure in Tianwan it was possible to detect and correct most of failures and non-conformance’s at early stages of engineering process. This ensures as well as high quality of safety system and efficiency of the entire engineering, installation and commissioning process.



Test Concept for Safety I&C System TELEPRM XS

Our experience from Tianwan I&C project confirms the importance of carefully and systematic testing. No any serious problem inside TXS safety system and in term of communication between TXS and TXP occurs.

4 CONCLUSION

FRAMATOME ANP, an AREVA and Siemens company, is a leading global provider of advanced nuclear technology as well as of advanced I&C solutions for nuclear power plants. During the past years, our innovative products and solutions have been chosen by numerous plant owners for new constructions as well as for major upgrades of their plant I&C.

FRAMATOME ANP is fully committed to continuously developing its TELEPERM platform to support the ever-increasing expectations for new and future nuclear plants. The automation of an entire nuclear power plant is a very challenging task. The unified platform TELEPERM XS/XP fulfills all automation tasks according to the requirements, rules and standards. The OM system offers a user-friendly HMI in normal and in accident situations including advanced diagnosis.